

AYSAN ESMAELY

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RESEARCH INTERESTS

Transportation equity and accessibility · Transportation policy analysis · Transportation and public health · Spatial analysis and GIS · AI in transportation

EDUCATION

M.Sc. in Community and Regional Planning

2024 – December 2025

University of Nebraska–Lincoln

Thesis: *Transportation Practitioners' Reflections on Addressing Transportation Barriers for Vulnerable Populations in Nebraska*

Advisor: Dr. Abigail Cochran

M.Sc. in Urban Planning (Coursework Completed)

2020 – 2022

Mazandaran University

Completed all required coursework; departure from Iran due to immigration circumstances prevented thesis defense. Transcript available upon request.

B.Sc. in Urbanism Engineering

2017

Payam Noor University

RESEARCH EXPERIENCE & PROJECTS

Graduate Research Assistant

University of Nebraska–Lincoln, 2024–2025

Southeast Nebraska Infrastructure Inventory | SENDD / Nebraska Dept. of Economic Development

- Co-designed and deployed a survey targeting city clerks and water operators across 15 counties.
- Served as primary contact for local officials; coordinated paper survey distribution and data entry.
- Participated in Evaluation & Selection Committee for Regional Planning Consultant RFP.

Transportation Barriers to Vision Care for the Visually Impaired | MATC-TSE / U.S. DOT

- Analyzed qualitative interview data from individuals with low vision; contributed to manuscript preparation.
- Conducted in-depth interviews with healthcare providers; performed thematic analysis for a separate manuscript.

Telehealth & Transportation Barriers in Rural Nebraska | SENDD / Infrastructure Investment and Jobs Act

- Independently designed study and conducted interviews with rural residents and healthcare providers.
- Analyzed qualitative data on broadband-enabled telehealth and transportation accessibility.

Studio Project

University of Nebraska–Lincoln, MCRP Studio, 2024–2025

City of Ogallala Comprehensive Plan Update |

- Led transportation section: background research, data collection, GIS mapping, analysis, and recommendations.
- Conducted community surveys, stakeholder interviews, and coordinated public open house at City Hall.

Thesis Research

University of Nebraska–Lincoln

2024 – 2025

"Transportation Practitioners' Reflections on Addressing Transportation Barriers for Vulnerable Populations in Nebraska"

- Conducted in-depth interviews with transportation specialists across Nebraska.
- Applied qualitative thematic analysis; thesis published on UNL Digital Commons (2025).

Mazandaran University, Iran

2020 – 2022

"Typology of Transportation Policies of Five-Year Central Government Comprehensive Plans and Their Effects on Tehran Transportation Plans"

- Reviewed national five-year planning documents to identify transportation policies and strategies.
- Analyzed how national-level policies were reflected in local planning documents for the City of Tehran.

PROFESSIONAL EXPERIENCE

Community Planner, Southeast Nebraska Development District (SENDN)

January 2026 – Present

Lincoln, Nebraska

- Authored and submitted the ECCBG grant for the City of Fairbury (~\$100,000), managing the full grant writing and submission process.
- Designed surveys and coordinated community data-gathering efforts to support municipal planning needs.
- Produced GIS maps and spatial analyses for NEPA environmental review, including project area delineation and proximity-to-airport assessments using ArcGIS.
- Supported environmental review processes and project scoping for infrastructure projects under NEPA compliance.
- Assisted communities with capital improvement planning, comprehensive plan updates, and stakeholder engagement.

Community Outreach Coordinator (Intern), SENDD

Summer 2024

Southeast Nebraska Development District

- Researched telehealth's role in overcoming transportation barriers in rural Nebraska.
- Contributed to the Capital Improvement Plan for the City of Waverly.
- Wrote planning analysis and recommendations for a Comprehensive Plan update for Pawnee City.

Planner Facilitator, Slum Upgrading Program

2017 – 2020

Gorgan, Golestan Province, Iran

- Led GIS-based mapping to update and formalize the city boundary line, incorporating informal settlement areas using ArcGIS.
- Built and maintained a comprehensive GIS database of informal settlement conditions for planning decisions.
- Conducted community engagement to identify and prioritize local challenges; developed multi-stakeholder action plans.
- Collaborated with a multidisciplinary team (economists, sociologists, engineers) on urban development projects.

Flood Disaster Volunteer

2019

Agh Ghala, Golestan Province, Iran

- Assessed critical infrastructure damage and mapped the most impacted areas.
- Projected needs of vulnerable populations to guide disaster response and aid distribution.

PUBLICATIONS

Published

Nelson-Forcade, C., **Esmaily, A.**, Shepherd, J.D., & Cochran, A.L. (2025). Clinicians' perspectives on transportation barriers to low vision care in the rural Midwest. *Findings*. <https://doi.org/10.32866/001c.129899>

Esmaily, A. (2025). *Transportation Practitioners' Reflections on Addressing Transportation Barriers for Vulnerable Populations in Nebraska*. M.Sc. Thesis. University of Nebraska–Lincoln. digitalcommons.unl.edu

Under Review

Esmaily, A., Nelson-Forcade, C., Shepherd, J.D., & Cochran, A.L. Addressing transportation barriers to low vision care among individuals with visual impairments in Nebraska. *Under review*.

In Preparation

Esmaily, A., & Cochran, A.L. Transportation practitioners' reflections on addressing transportation barriers for vulnerable populations in Nebraska.

Esmaily, A., Cochran, A.L., Bliss, T., & Borchers-Williams, O. The role of telehealth in overcoming transportation barriers in accessing healthcare in Nebraska.

Esmaily, A. AI agents in qualitative research: Examining machine-generated thematic analysis vs. human coding.

CONFERENCE PRESENTATIONS

Esmaily, A., Nelson-Forcade, C., Shepherd, J.D., & Cochran, A.L. Addressing transportation barriers to low vision care among individuals with visual impairments in Nebraska. Mar. 2025
Nebraska APA Chapter Conference

Esmaily, A., Nelson-Forcade, C., Shepherd, J.D., & Cochran, A.L. Addressing transportation barriers to low vision care. Nov. 2024
ACSP Annual Conference, Seattle, WA

Cochran, A.L., **Esmaily, A.**, Nelson-Forcade, C., & Choi, I. Understanding and addressing transportation barriers to low vision care in Nebraska. Oct. 2024
UNL Transportation Engineering Seminar Series, Lincoln, NE

TEACHING EXPERIENCE

Teaching Assistant — Urban Planning Workshop

2021 – 2022

Mazandaran University

- Assisted students with GIS-based projects in urban planning workshop for undergraduate Urbanism Engineering students.

GIS Instructor

2021

Gorgan, Iran

- Taught applied GIS skills including spatial analysis and mapping to professional learners.

TECHNICAL SKILLS

GIS & Spatial Analysis	ArcGIS (proficient), spatial analysis, boundary delineation, GIS database management, map production for NEPA/planning
Quantitative Analysis	SPSS, data visualization, survey design (Lasso, SurveyMonkey)
Qualitative Methods	Thematic coding, in-depth interviewing, Dedoose
Currently Learning	Python, machine learning, spatial modeling, GeoAI
Other	AutoCAD, grant writing, NEPA compliance, community engagement

AWARDS & HONORS

Student Scholarship Award for Outstanding Achievement — Nebraska APA Chapter	2025
Student Travel Award — American Planning Association (NPC, Denver)	2025
Graduate Student Travel Award — TRB Annual Conference, Washington D.C.	2025
Nebraska Transportation Center Award — TRB Annual Conference	2024
Graduate Student Travel Award — ACSP Annual Conference, Seattle	2024

PROFESSIONAL MEMBERSHIPS

American Planning Association (APA)
Association of Collegiate Schools of Planning (ACSP)
WTS: Advancing Women in Transportation
Student Planning Association of Nebraska (SPAN)

LANGUAGES

Turkmen: Native **Farsi:** Native **Turkish:** Fluent **English:** Advanced
